

INTRODUCTION – ROLE OF INFORMATION TECHNOLOGY IN LOGISTICS AND SUPPLY CHAIN MANAGEMENT

Information Technology (IT) has become an important part of business. It helps in **faster communication, quick processing of information and accurate performance of logistics activities**. It also helps in automation, which saves time and reduces errors.

1. Better Communication

IT helps in **quick sharing of information** between wholesalers, retailers, dealers and other members of the supply chain. This improves coordination and reduces mistakes.

2. Competitive Advantage

IT helps companies provide the **right product at the right time and place**. It also reduces costs, delays and errors, which helps the company perform better than competitors.

3. Ease of Operations

Tools like **EDI, LIS and ERP** help different departments share information easily. This makes the work **faster, smoother and more organised**.

4. Quick Decision Making

IT helps in **collecting, storing and analysing information quickly**. Managers can compare actual results with planned results and take better decisions.

5. Early Warning System

IT helps identify **problems or unusual situations at an early stage**. This allows the company to take corrective action before the problem becomes bigger.

6. Accurate Data

Technologies like **RFID and barcodes** help collect and store accurate information about goods. This helps in proper analysis and better decision-making.

7. Identifying Location of Consignment

Technologies like **GPS and satellite communication** help track the exact location of goods during transportation. This helps the company know when the goods are likely to reach the customer.

8. Other Benefits

IT provides faster processing and timely information. It helps generate useful reports, improves overall efficiency and reduces labour costs through automation.

TOOLS OF INFORMATION TECHNOLOGY IN LOGISTICS

The important IT tools used in logistics and supply chain management are:

1. EDI – Electronic Data Interchange

EDI helps companies **exchange business information electronically**. It makes communication faster and reduces paperwork and errors.

2. LIS – Logistics Information System

LIS helps in **collecting, storing and managing logistics information**. It helps the company manage activities like transportation, inventory and orders efficiently.

3. ERP – Enterprise Resource Planning

ERP connects **different departments of an organisation** and allows them to share information. It helps improve coordination and makes business operations smoother.

4. RFID – Radio Frequency Identification

RFID uses electronic tags to **identify and track goods**. It helps the company know the movement and status of products accurately.

5. Barcoding

Barcodes are used to **identify products and record their information quickly**. They reduce manual errors and make activities like billing, inventory checking and order processing easier.

6. GPS – Global Positioning System

GPS helps in **tracking the exact location of vehicles and consignments**. It helps companies monitor deliveries and estimate arrival time.

7. Satellite Communication

Satellite communication helps in **sending and receiving information over long distances**. It is useful for tracking and communicating with vehicles and consignments during transportation.

MODERN LOGISTICS INFRASTRUCTURE

Meaning: Modern logistics infrastructure refers to the facilities and systems developed to make the movement, storage and transportation of goods faster, safer and more efficient.

1. Golden Quadrilateral:

It is a network of highways connecting India's four major metropolitan cities – Delhi, Mumbai, Chennai and Kolkata. It helps in faster movement of goods and carries a major share of national traffic.

2. Logistics Parks: Logistics parks are locations where different logistics companies and service providers operate to

gether. They act as a one-stop place for various logistics needs and provide space for multiple clients.

3. Deep Water Ports:

Deep water ports are ports that can accommodate fully loaded large ships. They provide modern facilities for loading and unloading goods and support large-scale international trade.

4. Dedicated Freight Corridor:

It is a separate railway network specially used for freight trains. It reduces dependence on roads and helps in faster and more reliable transportation of goods.

5. Inland Container Depots (ICD):

ICDs are facilities used for temporary storage of import and export containers. They also provide facilities for loading, unloading and transportation of containers and help reduce congestion at ports.

6. Maritime Logistics:

Maritime logistics involves the transportation of goods and people through seas and other waterways. Containerisation has made transportation by sea faster and more efficient.

7. Double Stack Containers:

It is a system where two containers are stacked one above another on railway cars. It helps transport a large quantity of goods while saving time and energy.

8. Cold Chain Storage: Cold chain storage provides temperature-controlled facilities for products that need special temperatures. It includes cold storages, refrigerated vehicles and pre-cooling facilities and is mainly used for perishable goods.

OUTSOURCING

Meaning: Outsourcing means contracting one or more business activities of a company to an outside service provider. The outside agency performs the activity on behalf of the company. It helps the company reduce costs and focus on its main business activities.

REASONS FOR OUTSOURCING:

1. Reduction in Costs:

Outsourcing helps the company reduce operating costs because specialised outside agencies can perform the work at a lower cost.

2. Focus on Core Competencies:

The company can concentrate on its main business activities instead of spending time and resources on supporting activities.

3. Acquiring New Skills:

Outside service providers have specialised knowledge, skills and technology which may not be available within the company.

4. Better Management:

Outsourcing allows specialised agencies to manage particular activities efficiently, resulting in better performance.

5. Avoid Labour Problems:

The company can reduce problems related to recruitment, training, salaries and management of employees.

6. Focus on Business Policies and Strategies:

Management gets more time to concentrate on important business decisions, future plans and growth strategies.

7. Reduce Investment:

The company can avoid heavy investment in machines, technology, infrastructure and other resources required for the outsourced activity.

8. Flexibility:

Outsourcing provides flexibility to increase or decrease services according to the changing needs of the business.

9. Improve Overall Performance:

By using specialised service providers, the company can improve the quality, efficiency and speed of its operations.

TYPES OF 3PL SERVICE PROVIDERS

Meaning: 3PL (Third-Party Logistics) service providers are outside companies that perform logistics activities such as transportation, warehousing, packaging and distribution for another company.

1. Standard Service Provider:

These 3PL providers perform basic logistics activities such as **picking, packing, warehousing and distribution**. They mainly handle routine logistics work.

2. Service Developer: These providers offer **value-added services** along with basic logistics activities. Examples include **tracking and tracing, cross-docking and special packaging**.

3. Customer Adapter: A customer adapter works **at the request of the customer** and takes over the complete logistics activities of the company. It manages logistics operations on behalf of the company.

4. Customer Developer: This is the **highest level of 3PL service**. The provider becomes closely integrated with the company and takes responsibility for the **entire logistics operation** of the company.

IMPACT OF GLOBALISATION ON LOGISTICS & SUPPLY CHAIN MANAGEMENT

1. Emergence of New Markets:

Globalisation allows companies to enter new countries and markets. This gives them access to new customers, suppliers and resources.

2. New Suppliers:

Companies can purchase raw materials and products from suppliers located in different countries. This can provide more choices and better sourcing opportunities.

3. Collaborative Environment:

Globalisation encourages companies to work together with suppliers, distributors and logistics partners. This helps in reducing costs and focusing on core activities.

4. Competitive Pricing:

Companies face competition from businesses across the world. This encourages them to reduce logistics costs and provide products at competitive prices.

5. Changes in Supply and Demand:

Global markets make supply and demand more difficult to predict. Changes in one country can affect the supply chain of companies in other countries.

6. Increased Logistics Activities:

Globalisation increases the movement of goods between different countries. Therefore, transportation, warehousing and distribution activities become more important.

7. More Complex Supply Chain:

Managing suppliers, transportation and customers across different countries makes supply chain management more complex.

8. Increased Pressure on SCM:

Companies need faster delivery, better coordination, accurate information and efficient logistics to compete in the global market.

GLOBAL LOGISTICS TRENDS

1. Production Closer to End User:

Companies are bringing production units closer to customers. This reduces transportation distance, lead time and makes logistics planning easier.

2. Focus on Large Vessels:

Shipping companies are using larger ships to transport goods. Larger vessels can carry more goods and help reduce the transportation cost per unit.

3. Focus on Green Logistics:

Companies are focusing on reducing pollution and harmful emissions from transportation. More importance is given to environmentally friendly modes like railways.

4. Investment in IT Solutions:

Companies are increasingly using advanced IT systems to manage logistics. IT helps in tracking goods, improving communication, coordination and transparency among supply chain members.

5. Growth of E-Commerce:

Online shopping is increasing rapidly. This has increased the importance of last-mile delivery, where goods are delivered directly to customers or nearby distribution points.

6. Globalisation:

Companies are expanding into international and emerging markets. This increases the movement of goods between countries and expands the scope of logistics activities.

7. Ethical Supply Chain:

Customers are becoming more concerned about how products are manufactured and sourced. Companies therefore focus on reducing social, economic and environmental harm and following ethical practices.

8. Increasing Complexity of Supply Chains:

Supply chains are becoming more complex and dynamic. Sourcing locations keep changing and purchase orders are becoming smaller and more frequent, making supply chain management more challenging.

GLOBAL ISSUES AND CHALLENGES IN LOGISTICS & SUPPLY CHAIN MANAGEMENT

Introduction: Globalisation provides many opportunities to businesses, but operating at the international level also creates several challenges in logistics and supply chain management.

1. Increasing Fluctuation in Customer Demand:

Demand and supply can change frequently in the global market. Since customers are spread across different countries, it becomes difficult for companies to forecast demand accurately.

2. Increased Complexity in Supply Chain: A global supply chain involves many suppliers, customers, warehouses and production units located in different countries. Managing and controlling all these activities becomes difficult.

3. Political and Legal Differences: Every country has different political conditions, laws, rules and regulations. Companies have to understand and follow the requirements of each country in which they operate.

4. Cultural Differences: Different countries have different cultures, traditions, values and customer preferences. Companies need to understand these differences while managing their global operations.

5. Currency Differences: Different countries use different currencies. Changes in exchange rates and problems related to currency conversion can increase the cost and create difficulties in international transactions.

6. Language Differences: Language differences can create problems in communication between suppliers, employees, customers and logistics partners. Even the same words may have different meanings in different countries.

7. High Transportation Cost:

International markets are usually far from the production and supply locations. This increases transportation costs and also increases the time required for delivery.

Conclusion: Thus, global logistics requires proper planning and coordination to manage differences in demand, laws, culture, currency, language and transportation across countries.